Techniques in Stochastic Analysis

Clark Miyamoto (cm6627@nyu.edu)

May 25, 2026

Abstract

Some informal lectures notes on techniques in stochastic analysis.

Contents

I Generative Models **2**

1 Review of Stochastic Differential Equations **3**
1.1 Particle Trajectories & Probability Densities 4
1.2 Reversing Time 6
1.3 Tweedie's Formula & Score Based Diffusion 7

2 Stochastic Interpolants **9**
2.1 ODE Formulation 9
2.2 SDE Formulation 10

II Markov Semigroup Theory **12**

3 Discrete-Time Markov Chains **12**
3.1 Stationary Distribution 14
3.2 Convergence to Stationary Distribution 17

4 Continuous-Time Markov Chains **19**
4.1 Convergence to Stationary Distribution 20

Part I
Generative Models

Some lecture notes from when I TA'ed *Machine Learning for Physicists*. To your benefit (or dismay) the writing style is for a grad student in the physics department.

Resources Some reading material that I found helpful when learning about generative models.

- https://diffusion.csail.mit.edu/2026/index.html. Accessible and well motivated notes. Notes are typeset and have recorded lectures. Additionally has a section on **discrete diffusion**.
- https://alexxthiery.github.io/posts/reverse_and_tweedie/reverse_and_tweedie.html. A simple note on Tweedie's formula & reverse time SDEs.
- https://joanbruna.notion.site/Syllabus-667e0fa5659448cd861f3326dec873a0. Notes from Joan Bruna's course (lectures 9-12). More mathematically rigorous, but at the cost of accessibility.
- https://arxiv.org/pdf/2303.08797. The original Stochastic Interpolants paper which unifies ODE & SDE models. It's a great read, but assumes you're already familiar with generative models.
- https://lilianweng.github.io/posts/2021-07-11-diffusion-models/. Once you feel comfortable with the theory, the question becomes how do we implement these in practice. This blog gives a nice overview of problems & solutions in implementation.

The generative problem Generative models tackle the following problem

- **Given:** A dataset of samples $\mathcal{D} = \{x_i\}_i$, s.t. $x_i \overset{iid}{\sim} \pi$.
- **Objective:** Generate new samples from $x \sim \pi_\theta \approx \pi$. Where π_θ is implicitly defined by a neural network.

Note this is different from sampling; there you're given oracle access to $\pi(x)$, rather than samples.

The history of solving problem was originally solved by **(Variational) Autoencoders**. You learn an encoder & decoder, which maps you from $\mathsf{Data} \overset{E^\theta}{\to} \mathsf{Latent} \overset{D^\phi}{\to} \mathsf{Data}$; if you can easily sample Latent then plugging it into D^ϕ will sample Data. But why learn two models? What if Latent was something generic, like a Gaussian distribution, and I learn a deterministic map s.t. the pushforward maps it to Data. I'll denote this map as T_θ. Recall you can use the change of variables formula for probability distributions $p(x) = q(T^\theta(\cdot))|\det T^\theta|$, but you'll need $\det T_\theta$ to be computationally feasible. You can use some tricks to make this happen, these class of models are called **Normalizing Flows**. However (1) these class of models are notoriously difficult to train, (2) there are infinitely many T_θ which do the mapping. It would be nice if there was more structure. You can imagine decomposing the map $T_\theta = T_1^\theta \circ \ldots \circ T_n^\theta$, at each level it defines a probability distribution $(p_i^\theta)_{i\in[1,\ldots,n]}$ such that $p_1 = \mathsf{Data}$ and $p_n = \mathsf{Gaussian}$. What if we assert each probability distribution has to be a noisier version of the previous, until it's indistinguishable from a Gaussian, and instead we learn a denoising oracle. These are called **Diffusion** models. The training is a lot more stable, and we can start to make models that can sample high-dimensional distributions. However we have to specify the number of noising steps n and then train the model; I want to specify this after training. What if there was a continuous-time formulation of this process, and you can choose the time-discretization post-hoc? This is called **Score-Base Diffusion**. Effectively you learn a term in a stochastic differential equation (SDE). You realize that simulating the SDE is very expensive, and you want to instead simulate an ODE, but moves through the same family of probability distributions. These are called **Flow Matching** models. Finally, you think about uniting the two frameworks, allowing the user to specify a path in probability space and then choose wether to train an ODE or SDE. This is called **Stochastic Interpolants**. This is currently where the field is at.

In these lecture notes we'll cover Score-Base Diffusion and Stochastic Interpolants; which I find to be the most elegant & natural to the physicist.

1 Review of Stochastic Differential Equations

You've probably heard of SDEs before, but they aren't covered in the main-stream physics education. So I'll attempt to do a brief introduction.

There was a botanist studying pollen grains in water. He noticed the motion was jittery, moving randomly in all directions. You can imagine a heuristic model being

$$X_{t+h} = X_t + h^\alpha \, Z_t \tag{1.1}$$

where $Z_t \sim \mathcal{N}(0, \mathbb{I})$ (iid at every time t) is random noise, and h is the step size (according to the time-discretization) to the power α. In an attempt to find a continuous time model in the limit $h \to 0$ (discretization goes to zero), I'll recurse to time zero.

$$\begin{aligned} X_t &= X_{t-h} + h^\alpha \, Z_{t-h} && (1.2) \\ &= X_{t-2h} + h^\alpha \, (Z_{t-2h} + Z_{t-h}) && (1.3) \\ &= X_{t-3h} + h^\alpha \, (Z_{t-3h} + Z_{t-2h} + Z_{t-h}) && (1.4) \\ &= X_0 + h^\alpha \sum_{n=1}^{t/h} Z_{t-nh} && (1.5) \end{aligned}$$

Since we're physicists, we can choose a coordinate system s.t. the initial position $X_0 = 0$. We now note that

$$h^\alpha \sum_{n=1}^{t/h+1} Z_{t-nh} \sim \mathcal{N}\big((0, h^{2\alpha-1} t\big) \tag{1.6}$$

To keep the model independent on the size of the discretization, I'll choose $\alpha = 1/2$. Leaving us with

$$X_t - X_0 \sim \mathcal{N}(0, t) \tag{1.7}$$

This was quite heuristic, but we have some take aways. When making an infinitesimal that behaves randomly, it has units $\sqrt{t}$.

Now that we have some intuition for how this motion behaves, we can develop something more rigorous but has the same characteristics.

Definition 1 (Weiner Process / Brownian Motion)**.** Brownian motion $(W_t)_{t\geq 0}$ is a stochastic process such that

1. Initializes at zero: $W_0 = 0$
2. Normal increments: $W_t - W_s \sim \mathcal{N}(0, (t-s)\mathbb{I})$, for $0 \leq s \leq t$.
3. Independent increments: $W_{t_1} - W_{t_0}$ is independent from $W_{t_i} - W_{t_j}$.

The idea of a stochastic differential equations is to extend the dynamics of ODEs to the dynamics where you have random fluctuations of force. Such things are no-where differentiable, so how can we recover a derivative-esq operation w/o using a derivative? Well ODEs have that

$$\frac{dX_t}{dt} = \mu_t(X_t) \implies X_{t+h} = X_t + h\, u_t(X_t) + \mathcal{O}(h^2) \tag{1.8}$$

Similarly for an SDE (ODE with stochastic fluctuations) have a discretization

$$\begin{aligned} X_{t+h} &= X_t + hu_t(X_t) + (W_{t+h} - W_t)\,\sigma_t(X_t) + \mathcal{O}(h^{3/2}) && (1.9)\\ &= X_t + hu_t(X_t) + \sqrt{h}\sigma_t(X_t)Z + \mathcal{O}(h^{3/2}) && (1.10) \end{aligned}$$

This particular choice of discretization is called the **Euler-Maruyama Discretization**. The $\mathcal{O}(h^{3/2})$ is because $W_{t+h} - W_t$ is order $\sqrt{h}$, so you can have a cross term like $h(W_{t+h} - W_t)$. For brevity, we'll use a shorthand for 1.9

$$dX_t = \mu_t(X_t)\, dt + \sigma_t(X_t)\, dW_t \tag{1.11}$$

1.1 Particle Trajectories & Probability Densities

Think about simulating the trajectory of individual particles. Re-running the simulation (w/ different RNG seeds) gave slightly different trajectories due to dW_t, meaning there's implicitly a probability distribution at each point in time. Turns out there's an exact correspondence between the SDE governing particle's motion, and a PDE which is the probability distribution over particles.

Theorem 1 (Fokker-Planck Equation)**.** Consider the stochastic differential equation

$$\begin{aligned} dX_t &= \mu_t(X_t)\, dt + \sigma_t dW_t && (1.12)\\ X_0 &\sim p_0 \qquad \text{Boundary condition} && (1.13) \end{aligned}$$

where $\mu_t : [0,1] \times \mathbb{R}^d \to \mathbb{R}^d$ and $\sigma_t : [0,1] \to \mathbb{R}_{\geq 0}$ are deterministic functions. Then the corresponding probability distribution $X_t \sim p_t$ solves a partial differential equation of the following form

$$\begin{aligned} \partial_t p_t(x) &= -\nabla \cdot (\mu_t\, p_t) + \frac{\sigma_t^2}{2}\Delta p_t && (1.14)\\ p_{t=0} &= p_0 \qquad \text{Boundary condition} && (1.15) \end{aligned}$$

Proof: Since X_t is a random variable, it has a corresponding probability density function. I'll notate this as p_t. Now you need to show that p_t have a the corresponding time evolution. The trick to do this, is to recall the trick you employ when you show something is secretly a delta function. You would integrate it against a test function $f(x)$ and show it behaved as expected. We'll do the same thing.

$$\partial_t \mathbb{E}[f(X_t)] = \lim_{h\to 0} \frac{1}{h}\mathbb{E}[f(X_{t+h}) - f(X_t)] \tag{1.16}$$

$$= \lim_{h\to 0} \mathbb{E}[\nabla f^T\, u_t(X_t) + \frac{\sigma_t^2}{2}\Delta f(X_t) + \mathcal{O}(h)] \qquad \text{Taylor expand} \tag{1.17}$$

$$= \int \nabla f^T(x) u_t(x) p_t(x) + \frac{\sigma_t^2}{2}\Delta f(x) p_t(x)\, dx \tag{1.18}$$

$$= \int -f(x)\, \nabla\cdot(u_t(x)p_t(x)) + f(x)\frac{\sigma_t^2}{2}\Delta p_t(x)\, dx \quad \text{Integrate by parts} \tag{1.19}$$

On the LHS

$$\partial_t \mathbb{E}[f(X_t)] = \int f(x)\partial_t p_t(x)\, dx \tag{1.20}$$

Put LHS = RHS, and you're done. □

This is an extremely powerful thing you've just derived so a couple of notes

- Numerically, instead of solving a PDE, you can simulate it by running a bunch of particles via the SDE and histogramming them.
- The Greens function $G(x', t', x, t)$ correspond to transition amplitudes $p(x', t'|x, t)$. This is not so surprising because in QFT the greens function corresponded to tree-level 2-point functions (covariances), and the kernel in Gaussian Processes corresponded to the autocorrelation time (covariance across time).

Example: (Overdamped) Langevin Dynamics What choice of μ_t and σ_t leaves the distribution invariant $\partial_t p_t = 0$?

$$0 = \partial_t p_t = -\nabla\cdot(\mu_t p_t) + \frac{\sigma_t^2}{2}\Delta p_t \tag{1.21}$$

$$= \nabla\cdot(-\mu_t p_t + \frac{\sigma_t^2}{2}\nabla p_t) \tag{1.22}$$

$$= \nabla\cdot[(-\mu_t + \frac{\sigma_t^2}{2}\nabla\log p_t)p_t] \tag{1.23}$$

If you choose $\mu_t(x) = \frac{\sigma_t^2}{2}\nabla\log\pi(x)$, then the inside term vanishes ($\partial_t p_t = 0$) when $p_t = \pi$. The resulting SDE is

$$dX_t = \frac{\sigma_t^2}{2}\nabla\log\pi_t(X_t)\, dt + \sigma_t dW_t \tag{1.24}$$

Some comments

- In physics, $\sigma_t \propto \sqrt{kT}$ (where k is Boltzmann's constant, and T is temperature) and $p(x) = e^{-\beta U(x)}/Z$ (Boltzmann distribution). Implying $\nabla\log p_t(x) = -\beta\nabla U(x)$. Physically it is the equation of motion of a particle moving in a viscous fluid.
- In statistics, if we set $\log p_t = \log\pi$ (target distribution), we can use this as a sampling algorithm.
- It takes infinite time to converge to the stationary distribution. Time-discretization makes you sample a biased-distribution (which you can correct using a Metropolis Hastings step).

Example: Ornstein-Uhlenbeck Process (Variance Preserving) Let's solve the SDE where $\mu_t(X_t) = -\mu\, X_t$ and $\sigma_t = \sigma_t$. Note: "solution" to an SDE means we write $X_t = [...]$ as a random variable (or equivalently its distribution).

$$dX_t = -\mu\, X_t\, dt + \sigma_t dW_t, \qquad X_{t=0} = X_0 \tag{1.25}$$

- If we recall the Langevin equation, $\nabla \log p_t(x) = -\frac{\mu\sigma_t^2}{2}\, x$, we recover the OU process. What distribution has that as a gradient...? A Gaussian! So it will converge to a Gaussian in the infinite time-limit.

- Using Fokker Planck, we have a PDE on $\mathrm{Law}(X_t) = p_t$

  $$\partial_t p_t = \mu \nabla \cdot (x\, p_t) + \frac{\sigma_t^2}{2} \Delta p_t, \qquad p_{t=0} = \mathrm{Law}(X_0) := \pi \tag{1.26}$$

  You can get the Greens function by solving this in Fourier space. It is given by

  $$p(x, t|x_0) = \frac{1}{(2\pi\Sigma^2(t))^{d/2}} \exp\left(-\frac{|x - x_0 e^{-\mu t}|^2}{2\Sigma^2(t)}\right) \tag{1.27}$$

  where $\Sigma(t) = \int_0^t \sigma_s^2 e^{-2\mu(t-s)}\, ds$.

- This is the forward process of diffusion models. In particular, it is the Variance Preserving SDE.

Example: Heat Equation (Variance Exploding) In the case you noise the data by

$$dX_t = \sigma_t dW_t \tag{1.28}$$

this is called the variance exploding process. It is recommended to use the VP-process, but I've included it here for completeness.

1.2 Reversing Time

Theorem 2 (Reverse Kolmogorov Equation)**.** Given an SDE

$$dX_t = \mu_t(X_t)\, dt + \sigma_t\, dW_t, \qquad \mathrm{Law}(X_{t=0}) = p_0 \tag{1.29}$$

$$\partial_t p_t = -\nabla \cdot (\mu_t p_t) + \frac{\sigma_t^2}{2} \Delta(p_t) \qquad p_{t=0} = p_0 \tag{1.30}$$

which evolves forwards in time $t = 0, ..., T$. There is a corresponding reversed process, which runs in $\tau = T - t$ (where τ increments from $0, ..., T$)

$$dX_\tau = [-\mu_{T-\tau}(X_\tau) + \sigma_{T-\tau}^2 \nabla \log p_{T-\tau}(X_\tau)]\, d\tau + \sigma_{T-\tau} d\tilde{W}_\tau \qquad \mathrm{Law}(X_{\tau=0}) = q_0 \tag{1.31}$$

$$\partial_\tau q_\tau = -\nabla \cdot [(-\mu_{T-\tau}(x) + \sigma_{T-\tau}^2 \nabla \log p_{T-\tau}(x)) q_\tau] + \frac{\sigma_{T-\tau}^2}{2} \Delta q_\tau \qquad q_{\tau=0} = p_T \tag{1.32}$$

where $q_\tau = p_{T-\tau}$.

Proof: Let $\tau := T - t$ and $q_\tau := p_{T-\tau} = p_t$. Let's use change of basis to find the τ-time evolution on q_τ.

$$\begin{aligned} q(\tau) &= p(T-\tau) && (1.33)\\ \partial_\tau q(\tau) &= \partial_\tau q(T-\tau) && (1.34)\\ &= -q'(T-\tau) \quad \text{Chain Rule} && (1.35) \end{aligned}$$

Let's substitute in the Fokker Planck equation. For simplicity $f_\tau = \mu_{T-\tau}$ and $g_\tau = \sigma_{T-\tau}$

$$\begin{aligned} \partial_\tau q_\tau &= \nabla \cdot (f_\tau \, p_{T-\tau}) - \frac{g_\tau^2}{2}\Delta p_{T-\tau} && (1.36)\\ &= \nabla \cdot (f_\tau \, q_\tau) - \frac{g_\tau^2}{2}\Delta q_\tau \quad \text{Definition of } q_\tau && (1.37) \end{aligned}$$

In order to convert this back to the SDE formulation, the diffusion term must be positive semi-definite, so let's do some massaging

$$\begin{aligned} \partial_\tau q_\tau &= \nabla \cdot (f_\tau \, q_\tau) - \frac{g_\tau^2}{2}\nabla \cdot (\nabla q_\tau) && (1.38)\\ &= \nabla \cdot \left(f_\tau q_\tau - g_\tau^2 \nabla q_\tau\right) + \frac{g_\tau^2}{2}\nabla \cdot (\nabla q_\tau) && (1.39)\\ &= \nabla \cdot [(f_\tau - g_\tau^2 \nabla \log q_\tau) q_\tau] + \frac{g_\tau^2}{2}\Delta q_\tau && (1.40)\\ &= -\nabla \cdot [(-f_\tau + g_\tau^2 \nabla \log q_\tau) q_\tau] + \frac{g_\tau^2}{2}\Delta q_\tau && (1.41) \end{aligned}$$

This gives us the time-reverse Fokker Planck. Now we can just apply our correspondence between PDE and SDE to get the reverse-SDE. ☐

In diffusion models, if you use the reverse SDE at test-time, then this is the continuous-time formulation of diffusion models. This is called **score-based diffusion**.

1.3 Tweedie's Formula & Score Based Diffusion

Tweedie's Formula So the question is what is $\nabla \log p_t(x)$? So recall under the OU process, we say

$$X_t = \alpha_t X_0 + \beta_t Z \tag{1.42}$$

where $x_0 \sim \pi$ and $z \sim \mathcal{N}(0, \mathbb{I})$ (where they're drawn independently). Let $p_t := \text{Law}(x_t)$. Since you're adding two random variables, this corresponds to a convolution

$$p_t(x_t) = \int \pi(x_0) \frac{1}{Z_t} \exp\left(-\frac{(x_t - \alpha_t x_0)^2}{2\beta_t^2}\right) dx_0 \tag{1.43}$$

Turns out the score corresponds to a posterior expectation. This result is called **Tweedie's Formula**

$$\begin{aligned} \nabla \log p_t(x_t) &= \frac{1}{\int \pi(x_0)\exp\left(-\frac{(x_t - \alpha_t x_0)^2}{2\beta_t^2}\right)} \nabla_{x_t} \int \pi(x_0) \exp\left(-\frac{(x_t - \alpha_t x_0)^2}{2\beta_t^2}\right) && (1.44)\\ &= \frac{-1}{\dots} \int \pi(x_0) \exp(\dots) \frac{(x_t - \alpha_t x_0)}{\beta_t^2} \, dx_0 && (1.45)\\ &= -\mathbb{E}_{X_0|X_t}\left[\frac{X_t - \alpha_t X_0}{\beta_t^2} \middle| X_t = x_t\right] && (1.46)\\ &= -\frac{x_t - \alpha_t \, \mathbb{E}_{X_0|X_t}[X_0|X_t = x_t]}{\beta_t^2} && (1.47) \end{aligned}$$

where the expectation $\mathbb{E}_{X_0|X_t}$ is taken over the distribution $p(x_0|x_t) = \frac{1}{Z_t}\pi(x_0)\exp\left(-(x_t - \alpha_t x_0)^2/2\beta_t^2\right)$. An interpretation is the denoiser (best guess of clean data X_0 given noisy observation x_t) can be read from a rewritting of the equation: $\mathbb{E}[X_0|X_t = x_t] = (x_t + \beta_t^2 \nabla \log p_t(x_t))/\alpha_t$; saying the denoised guess, is the your observation plus moving it up the probability distribution a little.

Another interpretation comes from applying our defining relationship $X_t = \alpha_t X_0 + \beta_t Z$, to rewrite the score

$$\nabla \log p_t(x_t) = -\frac{1}{\beta_t}\mathbb{E}[Z|X_t = x_t] \tag{1.48}$$

These conditional expectations are nice, because they are the optimizer of the mean squared error. Meaning you can model them by minimizing the L_2 loss

$$\arg\min_f \mathbb{E}[(x - f(y))^2] = \mathbb{E}[x|y] \tag{1.49}$$

Proof:

$$\begin{aligned}
\mathbb{E}[(x - f(y))^2] &= \mathbb{E}[x^2 + f(y)^2 - 2xf(y)] && (1.50)\\
&= \mathbb{E}[f(y)^2 - 2xf(y)] + C && (1.51)\\
&= \mathbb{E}[f(y)^2 - 2\mathbb{E}[x|y]f(y)] + C \qquad \text{Tower property} && (1.52)\\
&= \mathbb{E}[f(y)^2 + \mathbb{E}[x|y]^2 - 2\mathbb{E}[x|y]f(y)] + C && (1.53)\\
&= \mathbb{E}[(\mathbb{E}[x|y] - f(y))^2] + C && (1.54)
\end{aligned}$$

This is a convex objective, so $\arg\min_f$ yields the optimal choice: $\mathbb{E}[x|y]$. □

This is all to say, you can rewrite conditional expectations as L_2 loss functions. So we've reduced this fancy problem to just minimizing the mean squared error

$$\begin{aligned}
\mathcal{L}(\theta) &= \mathbb{E}_{x_0,z,t}[(\eta_t^\theta(x_t) - z)^2] && (1.55)\\
&= \mathbb{E}_{x_0\sim\pi}\mathbb{E}_{t\sim\text{Unif}[0,T]}\mathbb{E}_{z\sim\mathcal{N}(0,\mathbb{I})}[(\eta_t^\theta(\alpha_t x_0 + \beta_t z) - z)^2] && (1.56)
\end{aligned}$$

Such that $\nabla \log p_t(x_t) = -\eta_t^\theta(x_t)/\beta_t$.

Score Based Diffusion We can finally put all our knowledge together to construct our first generative model.

1. **Learning:** The neural network implicitly models the score

$$\nabla \log p_t^\theta(x_t) = -\eta_t^\theta(x_t)/\beta_t, \qquad \eta_t^\theta(x_t) = \mathbb{E}[Z|X_t = x_t] \tag{1.57}$$

   where $X_t = \alpha_t X_0 + \beta_t Z$. The neural network can learn by minimizing the mean squared error

$$\mathcal{L}(\theta) = \mathbb{E}_{X_0\sim\pi}\mathbb{E}_{t\sim\text{Unif}[0,T]}\mathbb{E}_{Z\sim\mathcal{N}(0,\mathbb{I})}[(\eta_t^\theta(\alpha_t X_0 + \beta_t Z) - Z)^2] \tag{1.58}$$

   where π is the target distribution.

2. **Inference:** The probability path was mapping $p_0 = \pi$ to $p_T \approx \mathcal{N}(0, \mathbb{I})$ via the OU process. To generate samples we simulate the reverse Fokker Planck & plug in the learned score

$$dX_\tau = [-\mu_{T-\tau}(X_\tau) + \sigma_{T-\tau}^2 \nabla \log p_{T-\tau}^\theta(x)]d\tau + \sigma_{T-\tau}\, dW_\tau \tag{1.59}$$

2 Stochastic Interpolants

2.1 ODE Formulation

In score based diffusion, you were constrained to using the OU process as your measure transport. Notice there are many paths (in the space of probability distributions) to go from a base distribution (which was previously Gaussian) to a target distribution. For example consider the transport equation

$$\partial_t p_t(x) = -\nabla \cdot (\mathbf{b}_t(x)\, p_t(x)), \tag{2.1}$$
$$p_{t=0} = p_0 \tag{2.2}$$

at $p_{t=0}$ it is our base distribution, and you want to select the drift vector field $\mathbf{b}_t(x)$ which gives your target $p_{t=1} = p_1$. This choice is non-unique, as there are many paths the particles could take from p_0 to p_1. So perhaps we should choose a path which is "easy" for samples to be transported.

Let's be very ambitious. I'll declare a stochastic process

$$I_t = \alpha_t x_0 + \beta_t x_1 + \gamma_t z \tag{2.3}$$

and I want to find the drift field $\mathbf{b}_t$ that forces the samples to be equal in distribution to $\text{Law}(I_t) = p_t$. We call this the **stochastic interpolant**.

- The samples $(x_0, x_1) \sim \nu(x_0, x_1)$ are defined on a lifted measure s.t. it marginalizes appropriately

  $$\int \nu(x_0, x_1)\, dx_1 = p_0(x_0) \qquad \int \nu(x_0, x_1)\, dx_0 = p_1(x_1) \tag{2.4}$$

  An example is just making them independent $\nu(x_0, x_1) = p_0(x_0)p_1(x_1)$.

- The $z \sim \mathcal{N}(0, \mathbb{I})$ is a white noise term, so we'll make it $z \perp \nu(x_0, x_1)$.

- Since the probability distribution has boundary conditions $p_{t=0} = p_0$ and $p_{t=1} = p_1$. We'd hope that $I_{t=0} = x_0 \sim p_0$ and $I_{t=1} = x_1 \sim p_1$. This implies

  $$\alpha_{t=0} = \beta_{t=1} = 1, \quad \alpha_{t=1} = \beta_{t=0} = 0, \quad \gamma_{t=0} = \gamma_{t=1} = 0 \tag{2.5}$$

We can derive the drift field by solving the equation. A nice trick is to solve it in Fourier space

$$\partial_t \tilde{p}_t(k) = i\mathbf{k} \cdot \widetilde{\mathbf{b}_t p_t}(k) \tag{2.6}$$

where

$$\tilde{p}_t(k) = \int e^{ikI_t} p(I_t)\, dI_t \tag{2.7}$$
$$= \int \exp\left(ik(\alpha_t x_0 + \beta_t x_1 + \gamma_t\, z)\right) \nu(x_0, x_1)\, \mathcal{N}(z; 0, \mathbb{I})\, d(x_0, x_1)\, dz \tag{2.8}$$

$$\partial_t \tilde{p}_t(k) = ik \int \dot{I}_t e^{ikI_t} p(I_t) dI_t \tag{2.9}$$

$$= ik \int \dot{I}_t e^{ikI_t} d\nu(x_0, x_1)\, d\mu(z) \qquad p(I_t)dI_t = d\nu(x_0,x_1)d\mu(z) \tag{2.10}$$

$$= ik \int \dot{I}_t e^{ikx} \delta(I_t - x) d\nu(x_0, x_1)\, d\mu(z)\, dx \qquad \text{Insert identity} \tag{2.11}$$

$$= ik \int e^{ikx} \underbrace{\left(\int \dot{I}_t \frac{\delta(I_t - x) d\nu(x_0, x_1)}{p_t(x)}\, d\mu(z) \right)}_{= \mathbb{E}[\dot{I}_t | I_t = x]} p_t(x) dx \tag{2.12}$$

where $p_t(x) = \int \delta(I_t - x) d\nu(x_0, x_1) d\mu(z)$. This is the distribution $p(I_t)$ in disguise, you can see this by integrating against a test function. You can now see you satisfy the transport equation in Fourier space, therefore the drift field which satisfies $\text{Law}(I_t) = p_t$ is given by

$$\boxed{b_t(x) = \mathbb{E}[\dot{I}_t | I_t = x]} \tag{2.13}$$

Training Now that we've specified our choice of stochastic interpolant, we should create a training goal. We'll use a neural network to approximate $\hat{b}_t(x) \approx b_t(x)$. A simple loss is mean-square error weighted by the probability distribution

$$\mathcal{L}(\hat{b}) = \int_0^1 \mathbb{E}[|\hat{b}_t(I_t) - \dot{I}_t^2|^2]\, dt \tag{2.14}$$

After you've converged, you can plug in your estimate into a discretization of 2.16. Notice this expectation is just over p_0, p_1, t; which in the generative problem you have access to.

Inference Because of the Fokker-Planck equation, I know the correspondance between the PDE on the distribution and the ODE on the samples.

$$\partial p_t(x) = -\nabla \cdot (b_t(x)\, p_t(x)), \;\; p_{t=0} = p_0 \tag{2.15}$$

$$\iff \frac{d}{dt} X_t = b_t(X_t), \;\; X_0 \sim p_0 \tag{2.16}$$

So now I can transport samples by specifying a choice of a stochastic interpolants I_t. Computing $b_t(x)$ is difficult, as it implictly depends on x_0, x_1, z, so perhaps we can learn it.

2.2 SDE Formulation

To recap, you've figured out how to learn an ODE which has $\text{Law}(I_t) = p_t$. Perhaps we can extend these to SDEs. Consider the Fokker-Planck equation

$$\partial_t p(x) = -\nabla \cdot (b_t^F(x)\, p_t(x)) + \epsilon_t \Delta p_t(x) \tag{2.17}$$

what is the b_t^F and ϵ_t that cause $\text{Law}(I_t) = p_t$? Instead of doing a bunch of Fourier stuff again, we can attempt to relate this current equation to the transport equation. Consider a decomposition of $b_t^F = b_t + f_t$, where $b_t(x) = \mathbb{E}[\dot{I}_t | I_t = x]$ is the drift from previously. Then

$$\partial_t p = -\nabla \cdot (b_t\, p_t) - \nabla \cdot (f_t\, p_t) + \epsilon_t \Delta p_t \tag{2.18}$$

$$\partial_t p = \partial_t p_t - \nabla \cdot (f_t\, p_t) + \epsilon_t \Delta p_t \tag{2.19}$$

$$0 = -\nabla \cdot (f_t\, p_t) + \epsilon_t \Delta p_t \tag{2.20}$$

By allowing the LHS and RHS $\partial_t p_t$ to cancel, you're implicitly saying the probability distribution that solved the transport also solves this Fokker Planck. Therefore any solution has $\text{Law}(I_t) = p_t$. Now we just have to choose f_t to get rid of the other term, this ends up being $f_t = \epsilon_t \nabla_x \log p_t(x)$

$$
\begin{aligned}
0 &= -\epsilon_t \nabla \cdot (\nabla_x \log p_t \, p_t) + \epsilon_t \Delta p_t && (2.21)\\
&= -\epsilon_t \Delta p_t + \epsilon_t \Delta p_t = 0 && (2.22)
\end{aligned}
$$

Therefore

$$\boxed{b_t^F(x) = b_t(x) + \epsilon_t \nabla_x \log p_t(x)} \tag{2.23}$$

The question now is what is the score? Because $\text{Law}(I_t) = p_t$ and we want $\nabla_x \log p_t(x)$ we should inspect the density for I_t. Consider the conditional

$$
\begin{aligned}
p_t(x) &= (\rho_t * \mathcal{N}(0, \gamma_t^2 \mathbb{I}))(x) && (2.24)\\
&= \int \rho_t(a) \frac{\exp(-(x-a)^2/2\gamma_t^2)}{Z_t} \, da && (2.25)\\
\nabla_x \log p_t(x) &= \frac{\nabla_x p_t(x)}{p_t(x)} && (2.26)\\
&= \frac{1}{p_t(x)} \int -\frac{(x-a)}{\gamma_t^2} \rho_t(a) \frac{\exp(-(x-a)^2/2\gamma_t^2)}{Z_t} \, da && (2.27)\\
&= \frac{1}{\int \rho_t(a) \frac{\exp(-(x-a)^2/2\gamma_t^2)}{Z_t} \, da} \int -\frac{(x-a)}{\gamma_t^2} \rho_t(a) \frac{\exp(-(x-a)^2/2\gamma_t^2)}{Z_t} \, da && (2.28)
\end{aligned}
$$

Think about what you've written down here, $\rho_t = \text{Law}(I(t, x_0, x_1))$ and $\mathcal{N}(0, \gamma_t^2 \mathbb{I}) = \text{Law}(\gamma_t z)$. If $\mathcal{N}(0, \gamma_t^2 \mathbb{I})$ is the likelihood and ρ_t is the prior, by Bayes theorem, this is an expectation over the posterior.

$$
\begin{aligned}
&= \frac{\mathbb{E}[I(t, x_0, x_1) | I_t = x] - x}{\gamma_t^2} && (2.29)\\
&= \frac{\mathbb{E}[I(t, x_0, x_1) - I_t | I_t = x]}{\gamma_t^2} && (2.30)
\end{aligned}
$$

$$\boxed{\nabla_x \log p_t(x) = -\frac{\mathbb{E}[z | I_t = x]}{\gamma_t}} \qquad \text{By def } I_t = I(t, x_0, x_1) + \gamma_t z \tag{2.31}$$

Directly the score is not very viable, so we'll approximate it with a neural network. Due to $\gamma_t \to 0$ at $t = 0, 1$, it'll be a numerically unstable quantity. Instead we'll instead learn just $\mathbb{E}[z | I_t = x]$; we'll denote this as the **denoiser**.

$$\boxed{\eta(x) = \mathbb{E}[z | I_t = x]} \tag{2.32}$$

To train the model, you once again use the fact that the conditional expectation is the minimizer in MSE.

Inference The learned drift & denoiser satisfy the Fokker-Planck equation

$$
\begin{aligned}
&\partial_t p = -\nabla \cdot (b_t^F \, p_t) + \epsilon_t \Delta p_t && (2.33)\\
\iff \; & dX_t = b_t^F(X_t) \, dt + \sqrt{2\epsilon_t} \, dW_t && (2.34)\\
&= \left[b_t(X_t) - \frac{\epsilon_t}{\gamma_t} \eta_t(X_t) \right] dt + \sqrt{2\epsilon_t} \, dW_t && (2.35)
\end{aligned}
$$

So if you have access to samples $X_0 \sim p_0$, then you can sample p_1. Additionally, you can tune ϵ_t after training to optimize performance.

Part II
Markov Semigroup Theory

The Sampling Problem In the previous lectures, we were concerned with the problem in generative models. You have access to examples $\{x_i\}_i \overset{iid}{\sim} \pi$, and you want to learn a model $\pi_\theta \approx \pi$ to generate new samples.

In these lectures we're concerned about a similar, yet very different problem. The sampling problem. Here you have oracle (point-wise) access to $\pi(x)$, and you want to generate samples $x \sim \pi$. Sometimes you only have access to the un-normalized density $\pi = \tilde{\pi}/Z$, where $Z = \int \tilde{\pi}\, dx$.

Motivation for learning this In the previous set of lectures, we briefly mentioned **Langevin dynamics**. Given a target distribution π, you can initialize your samples anywhere X_0 then time-evolve them according to

$$X_{t+h} = X_t + h\nabla \log \pi(X_t) + \sqrt{2h}\, Z \qquad \text{Discrete Time} \qquad (2.36)$$
$$dX_t = \nabla \log \pi(X_t)\, dt + \sqrt{2}\, dW_t \qquad \text{Continuous Time} \qquad (2.37)$$

and as $t \to \infty$, your samples will converge to $X_\infty \sim \pi$ (target measure).

Langevin fell under a special class of dynamics called **Markov Processes / Markov Chains**. These lectures will walk us through some of the tools & techniques to analyze these processes. Markov chains show us in many disciplines, so to give the reader a consistent viewpoint, I'll adopt my POV (which is a computational statistician interested in the math; meaning you only have oracle access to π, and sometimes only have oracle access to the normalize density $\tilde{\pi}$ associated with $\pi = \tilde{\pi}/Z$.), but hopefully the lessons can give the reader enough to generalize this knowledge to other problems.

From this view point, there's a couple questions that jump out to me

1. Can we develop other algorithms which can generate samples from π?
2. Once we specify a sampling algorithm, can we quantify how quickly samples will converge to π?
3. Many algorithms are easy to specify in continuous time, but we implement only in discrete time. Can we quantify how the discretization affects the analysis?

To keep us grounded, we'll work through these questions by looking back to Langevin, but the theory will stay general.

3 Discrete-Time Markov Chains

Consider a stochastic process $(X_t)_t$, and it's corresponding marginal over time $\mathrm{Law}(X_t) := \pi_t$. The name of the game is to track how a state π_t evolves over time. In these lecture notes, we'll look at a specific class of dynamics, called Markov Chains

Definition 2 (Time-Homogenous Markov Chain)**.** A discrete-time stochastic process $(X_t)_t$ is a Markov Chain if $p(X_{t+1}|X_t, ..., X_0) = p(X_{t+1}|X_t)$, denoted as the transition kernel. It is time-homogenous because the kernel is time-independent.

In this case, the dynamics of the system are completely governed by the transition kernel.

- When the state space Ω is continuous, π is a probability density function ($\pi \geq 0$ and $\int_\Omega \pi(x)\,dx = 1$). The dynamics are

$$\pi_{t+1}(y) = \int P(x,y)\,\pi_t(x)\,dx, \qquad P(x,y) = p(X_{t+1} = y|X_t = x) \tag{3.1}$$

  The notation of $P(x,y)$ is chosen s.t. it can be read, the probability of starting at x and ending up at y.

- If the state space is discrete, then π_t is a vector in the probability simplex (π_t is a vector, $[\pi_t]_i \geq 0$ and $\sum_i [\pi_t]_i = 1$). The dynamics turn into a matrix equation

$$\pi_{t+1} = \pi_t P, \qquad P_{ij} = \mathbb{P}(X_{t+1} = j|X_t = i) \tag{3.2}$$

  Because P_{ij} represents probabilities $\sum_j P_{ij} = 1$. This is called a **stochastic matrix**

Some other fun facts,

- When you apply P multiple times, you're evolving multiple steps. E.g. P^k time evolves for k steps.
- The eigenvector $\pi = \pi P$ corresponds to the **stationary distribution** / fixed point.

Example. Looking back at our motivating example, Langevin dynamics. It is also a Markov Chain

$$dX_t = \nabla \log \pi(X_t)dt + \sqrt{2}dW_t \tag{3.3}$$
$$X_{t+h} = X_t + h\,\nabla \log \pi(X_t) + \sqrt{2h}Z \qquad \text{(Discretization)} \tag{3.4}$$

Now let's derive the transition kernel $p(X_{t+h}|X_t = x)$

$$X_{t+h} = X_t + h\,\nabla \log \pi(X_t) + \sqrt{2h}Z \tag{3.5}$$
$$X_{t+h}|X_t = x \overset{d}{=} x + h\nabla \log \pi(x) + \sqrt{2h}Z \tag{3.6}$$

The probability distribution of Z is $\mathcal{N}(0, \mathbb{I})$. We can use the rules of Gaussian random variables

$$b + \sigma Z \sim \mathcal{N}(b, \sigma^2\mathbb{I}) \tag{3.7}$$
$$\therefore x + h\nabla \log \pi(x) + \sqrt{2h} \sim \mathcal{N}(x + h\nabla \log \pi(x),\ 2h\mathbb{I}) \tag{3.8}$$

So if the RHS is drawn from a Gaussian, therefore the LHS is also a Gaussian. Therefore the probability distribution of $X_{t+h} = x'|X_t = x$ is

$$p(x'|x) = \frac{1}{Z}\exp\Big(-(x' - (x + h\nabla \log \pi(x)))^2/4h\Big) \tag{3.9}$$

3.1 Stationary Distribution

Since our objective is to construct a Markov chains which has a stationary distribution. It turns out a chain has a stationary distribution under two sufficient conditions

1. *Irreducibility:* It is possible to get from x to another state x' w/ probability > 0 in a finite number of steps
2. *Aperiodicity:* It is possible to return to any state at any time. I.e. $\exists n$ s.t. $\forall \omega \in \Omega$ and $\forall n' > n$, $\mathbb{P}(X_{n'} = i | X_0 = i) > 0$.

The first prevents absorbing states, e.g. states that we can never leave. The second prevents transition operations such as

$$T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{3.10}$$

which is a chain that alternates between 1 and 2 without ever settling. These conditions constrain the types of Markov chains to get a nice result.

Theorem 3 (Perron-Frobenius)**.** An irreducible & aperiodic Markov Chain P

1. Admits a unique stationary distribution π
2. The stationary distribution π is the eigenvector of P corresponding to the largest eigenvalue $|\lambda| = 1$.

The proof is more about linear algebra than Markov Chains, so it's omitted.

The importance of the theorem is that the eigenvector corresponding to the largest eigenvalue is the stationary distribution. Visualize the coordinate transformation of the matrix P. Repeated applications of P will slowly project your vector along eigenvector

Fact. For an irreducible & aperiodic Markov Chain P with stationary distribution π, repeated applications of P will make any initialization π_0 converge to π.

$$\lim_{k \to \infty} \pi_0 P^k = \pi \tag{3.11}$$

The proof is more informative to convergence rates, so I'll prove it later.k

Detailed Balance and Metropolis-Hastings Algorithm You might wonder, given oracle access to π, how can we construct a Markov chain P which emits π as the stationary distribution.

Well, say you're at the stationary distribution π, then you'd hope for the following properties

1. Invariant under time-evolution: $\pi = \pi P$
2. Invariant under reverse time-evolution $\pi_i = \sum_j p(x_{t+1} = j | x_t = i)\pi_j \implies \pi = P^T \pi$

Definition 3 (Detailed Balance)**.** It is the balance between transition rates

$$\pi(y)p(x|y) = \pi(x)p(y|x) \tag{3.12}$$

where p is the transition kernel of a Markov Chain.

Fact. Let p be the transition kernel associated with a Markov chain. If (p, π) satisfies detailed balance, then π is the stationary distribution of the chain.

Proof.

$$\pi(y)p(x|y) = \pi(x)p(y|x) \tag{3.13}$$

Just integrate w.r.t. y

$$\sum_y \pi(y)p(x|y) = \pi(x)\sum_y p(y|x) \tag{3.14}$$

$$\sum_y \pi(y)p(x|y) = \pi(x) \qquad \sum_y p(y|x) = 1 \tag{3.15}$$

$$\pi P = \pi \tag{3.16}$$

and then x

$$\pi(y)\sum_x p(x|y) = \sum_x \pi(x)p(y|x) \tag{3.17}$$

$$\pi = \pi P \tag{3.18}$$

"It seems too easy", Jonathan Goodman. □

As the person running experiments, I simply want to specify a target distribution π & a proposal kernel q, and check which one works well. However not every transition kernel satisfies detailed balance for arbitrary π. Perhaps we can create a framework which always emits detailed balance.

Consider choosing a new kernel $p(x'|x) = q(x'|x)\,\alpha(x'|x)$, where I'll soon specify α. In the detailed balance equation

$$\pi(x)q(y|x)\alpha(y|x) = \pi(y)q(x|y)\alpha(x|y) \implies \alpha(y|x) = \frac{\pi(y)q(x|y)}{\pi(x)q(y|x)} \tag{3.19}$$

For this to be a probability, the ratio must be ≤ 1. However, this could easily be the other case, so let's symmetrize

$$\alpha(y|x) = \min\left(\frac{\pi(y)q(x|y)}{\pi(x)q(y|x)},\ 1\right) \tag{3.20}$$

Fact. The Metropolis Hasting algorithm satisfies detailed balance.

Algorithm 1 Metropolis–Hastings

Require: Target π, proposal $q(\cdot \mid x)$, initial X_0, steps N

for $t = 0, 1, \ldots, N-1$ **do**
  Propose $X' \sim p(\cdot \mid X_t)$
  $\alpha \leftarrow \min\left(1,\ \dfrac{\pi(X')\, p(X_t \mid X')}{\pi(X_t)\, p(X' \mid X_t)}\right)$
  Draw $u \sim \text{Uniform}(0,1)$
  if $u \leq \alpha$ **then**
    $X_{t+1} \leftarrow X'$ ▷ accept
  else
    $X_{t+1} \leftarrow X_t$ ▷ reject
  end if
end for
return $(X_0, \ldots, X_N)$

Proof. The corresponding transition kernel is

$$p(x|y) = q(x|y)\alpha(x|y) + \delta(x-y)R(y), \quad R(y) = \int q(x|y)(1-\alpha(x|y))dxd \tag{3.21}$$

Either $\alpha(y|x) = 1$ or $\alpha(x|y) = 1$. Assume the former

$$\begin{aligned}
\pi(y)p(x|y) &= \pi(y)[q(x|y) + \delta(x-y)R(y)] && (3.22)\\
&= \pi(y)q(x|y) + \pi(x)\delta(x-y)R(x) && (3.23)\\
&= \pi(x)q(y|x)\frac{\pi(y)q(x|y)}{\pi(x)q(y|x)} + \pi(x)\delta(x-y)R(x) && (3.24)\\
&= \pi(x)[q(y|x)\alpha(y|x) + \delta(x-y)R(x)] && (3.25)\\
&= \pi(x)p(y|x) && (3.26)
\end{aligned}$$

□

Example (Metropolis-Adjusted Langevin (MALA))**.** For Langevin dynamics, we proved that it has the stationary distribution π. However this was in continuous time. Does this still hold when we discretized time?

Recall the Euler-Maruyama discretized Langevin

$$\begin{aligned}
X_{t+h} &= X_t + h\nabla \log \pi(X_t) + \sqrt{2h}\, Z && (3.27)\\
p(x'|x) &= \frac{1}{Z}\exp\left(-\frac{(x' - x - h\nabla \log \pi(x))^2}{4h}\right) && (3.28)
\end{aligned}$$

this is known as **Unadjusted Langevin (ULA)**. It's clear the kernel doesn't satisfy detailed balance (for arbitrary π), as there's no π term in the kernel. Although detailed balance is "sufficient, not necessary" that you converge to π, turns out the discretization makes you sample a biased version of π. We can correct for this bias by using the proposal kernel in the Metropolis-Hasting step.

Let's calculate the acceptance ratio for the ULA kernel. This procedure is called **Metropolis-**

Adjusted Langevin (MALA)

$$\log \frac{p(x'|x)}{p(x|x')} = \frac{1}{4h}\left[(x' - x + h\nabla \log \pi(x))^2 - (x - x' + h\nabla \log \pi(x'))^2\right] \tag{3.29}$$

$$= \frac{h}{4}(\nabla \log \pi(x)^2 - \nabla \log \pi(x')^2) + \frac{1}{2}\langle x' - x, \nabla \log \pi(x) + \nabla \log \pi(x')\rangle \tag{3.30}$$

which you plug into $\alpha = \min\left(\frac{\pi(x)p(x'|x)}{\pi(x')p(x|x')},\ 1\right)$.

Problem 1. The previous MALA algorithm was stated for single step. Consider a modification to the MALA algorithm for multiple steps. What is the new correction factor $p(x'|x)/p(x|x')$ in the Metropolis acceptance ratio?

Sampling without Detailed Balance For example: sequential monte carlo & diffusion models, bouncy particle sampler.

3.2 Convergence to Stationary Distribution

It's nice that we can construct Markov chains where we know the stationary distribution. However, I need to run this and get some results, so perhaps we can engineer transition kernels with quicker convergence?

Discrete Space & Discrete Time Markov Chains Let's now prove convergence fact from the previous section, the proof will give us a nice insight into convergence rates.

Fact. Consider an irreducible & aperiodic Markov Chain P with stationary distribution π. Repeated applications of P will make any initialization π_0 converge to π.

$$\lim_{k\to\infty} \pi_0 P^k = \pi \tag{3.31}$$

Proof. For simplicity (and w/o loss of generality), assume P is diagonalizable with eigensystem $\{(\lambda_i, v_i)\}_{i=1}^d$, where $\lambda_1 > \ldots > \lambda_d$. By Perron-Frobenius theorem $\lambda_1 = 1$ and $v_1 = \pi$.

Decompose π_0 into a linear combination of the eigensystem

$$\pi_0 P^k = (c_1\pi + c_2 v_2 \ldots + c_d v_d)P^k \tag{3.32}$$

$$= c_1(1)^k \pi + c_2\lambda_2^k v_2 + \ldots + c_d\lambda_d^k v_d \tag{3.33}$$

$$\lim_{k\to\infty} \pi_0 P^k = c_1\pi + \mathcal{O}(\lambda_2^k) = c_1\pi \tag{3.34}$$

You can kill off λ_2 (and below) because it's smaller than $\lambda_2 = 1$. Because the state still lives in $\mathcal{P}(\Omega)$, it must normalize, therefore $c_1 = 1$.

Note: power iteration can be generically applied to find the eigenvector of the largest eigenvalue. The convergence rate is generically controlled by the **spectral gap** $|\lambda_1| - |\lambda_2|$. □

Notice, for finite k you're converging to the stationary distribution in corrections of λ_2 (second largest eigenvalue). This means spectral properties of the matrix controls the convergence rate of our Markov Chains.

Reversibility and Spectral Theorem The proof above assumed P is diagonalizable.

Fact (Detailed balance $\iff$ self-adjointness)**.** Let P be the transition matrix of a Markov chain with stationary distribution π, and suppose (P, π) satisfies detailed balance $\pi(x)P(x, y) = \pi(y)P(y, x)$. Then P is self-adjoint on the weighted space $L^2(\pi)$ equipped with the inner product

$$\langle f, g\rangle_\pi = \sum_x \pi(x)f(x)g(x). \tag{3.35}$$

Consequently P has real eigenvalues $1 = \lambda_1 > \lambda_2 \geq \cdots \geq \lambda_d \geq -1$ and a basis of eigenvectors orthonormal in $\langle\cdot,\cdot\rangle_\pi$.

Proof. Define $D = \text{diag}(\pi)$. Detailed balance reads $DP = (DP)^T = P^T D$, i.e. DP is symmetric. Form the similarity transform $S = D^{1/2}PD^{-1/2}$. Note it is symmetric

$$\begin{aligned} S^T &= D^{-1/2}P^T D^{1/2} && (3.36)\\ &= D^{-1/2}(DPD^{-1})D^{1/2} && P^T D = DP \quad (3.37)\\ &= D^{1/2}PD^{-1/2} = S, && (3.38)\end{aligned}$$

By the spectral theorem it has real eigenvalues and an orthonormal eigenbasis. Since $P = D^{-1/2}SD^{1/2}$ is similar to S, it shares the same (real) eigenvalues, and its eigenvectors are the columns of $D^{-1/2}$ applied to those of S, which are precisely orthonormal under $\langle\cdot,\cdot\rangle_\pi$. □

Notice

Definition 4 (Spectral Gap)**.** The spectral gap for a matrix P is defined as

$$\gamma = |\lambda_1| - |\lambda_2| \tag{3.39}$$

For irreducible & aperiodic Markov chains, $\lambda_1 = 1$, meaning the convergence is entirely controlled by λ_2.

Theorem 4. For a time-discrete & reversible Markov chain P with spectral gap $\gamma = 1-\lambda_2$ (2nd largest eigenvalue of P). We converge

$$\|\pi_k - \pi\|_{\mathsf{TV}} \leq \frac{1}{2}\sqrt{\chi^2(\pi_0\|\pi)}(1-\gamma)^k \tag{3.40}$$

where $\chi^2(\pi_0\|\pi) \equiv \int[(\pi_0(x)/\pi(x))^2 - 1]dx$

Continuous Space & Discrete/Continuous Time Markov Chains When the state space is continuous, the time-evolution P gets promoted to an operator. And you can imagine finding eigenvalues of some operator will be quite difficult. We need to change our tools a little bit!

4 Continuous-Time Markov Chains

When we go to continuous space Markov Chains, it's easy to start with continuous time and use the theory of ODE/PDEs to make statements, then bound the discretization error.

Definition 5 (Markov Semigroup Operator)**.** Given a Markov Process $(X_t)_{t\geq 0}$, tis semigroup $(P_t)_{t\geq 0}$ acts on test functions $f : \mathbb{R}^d \to \mathbb{R}$ by

$$(P_t f)(x) \equiv \mathbb{E}[f(X_t)|X_0 = x] \tag{4.1}$$

Why is it called a semigroup? In math a semigroup is when you have closure under composition $g_1 \circ g_2 \in G$, and have an identity $e \circ g = g \circ e \in G$ (s.t. $e \in G$). If you have closure under inverse, then this is promoted to be a group.

Property 1. Let's figure out in what sense this is a semigroup

1. Identity: $P_0 = \mathsf{Id}$.

   Proof: $(P_0 f)(x) = \mathbb{E}[f(X_0)|X_0 = x] = f(x)$, so P_0 acts as an identity operator.

2. Composition: $P_t \circ P_s = P_{t+s}$.

   Proof:

$$\begin{aligned}
(P_s f)(x) &= \mathbb{E}[f(X_s)|X_0 = x] && (4.2)\\
(P_t P_s f)(x) &= \mathbb{E}[\mathbb{E}[f(X_s)|X_0 = X_t]|X_0 = x] && (4.3)\\
&= \mathbb{E}[\mathbb{E}[f(X_{s+t})|X_t]|X_0 = x] && (4.4)\\
&= \mathbb{E}[f(X_{s+t})|X_0 = x] = (P_{t+s} f)(x) && (4.5)
\end{aligned}$$

Definition 6 (Generator)**.** The generator $\mathcal{L}$ is defined as

$$\mathcal{L} f := \lim_{h \downarrow 0} \frac{P_h f - f}{h} \tag{4.6}$$

Example. Let's compute the infinitesimal generator for Langevin dynamics. Since we'll take the time-increment to be small, we can start at the Euler-Maruyama discretization.

$$X_h = X_0 + h\nabla \log \pi(X_0) + \sqrt{2h} Z \tag{4.7}$$

where $Z \sim \mathcal{N}(0, \mathbb{I})$. To compute $P_h f = \mathbb{E}[f(X_h)|X_0 = h]$, we can apply a test function f to X_h, and then Taylor expand to second order, that way remaining terms are order h (taking the limit $h \to 0$ in the infinitesimal generator will kill $h^{3/2}$ terms).

$$\begin{aligned}
f(X_h) &= f(X_0) + f'(X_0)(X_h - X_0) + \frac{1}{2} f''(X_0)(X_h - X_0)^2 + \mathcal{O}((X_h - X_0)^3) && (4.8)\\
&= f(X_0) + \langle \nabla f(X_0), h\nabla \log \pi(X_0) + \sqrt{2h} Z \rangle \\
&\quad + \frac{1}{2}(h\nabla \log \pi(X_0) + \sqrt{2h} Z)^T \nabla^2 f(X_0)(-h\nabla \log \pi(X_0) + \sqrt{2h} Z) && (4.9)
\end{aligned}$$

Now we can take the expectation value

$$\mathbb{E}[f(X_h)|X_0 = x] = -h\langle \nabla f(x), \nabla \log \pi(x)\rangle + h\,\mathrm{Tr}[\nabla^2 f(x)] \tag{4.10}$$

Plugging into the definition of the generator

$$\mathcal{L}f = \langle \nabla f, \nabla \log \pi\rangle + \Delta f \tag{4.11}$$

Problem 2. Find the infintesimal generator associated for a generic stochastic process

$$dX_t = \mu_t(X_t)dt + \sigma_t dW_t \tag{4.12}$$

Does it look familiar?

4.1 Convergence to Stationary Distribution

To quantify the convergence rate to the stationary distribution, we could inspect a "distance" measure between the marginal p_t of the Markov Chain and the target π, and see if its growing / shrinking in time.

Definition 7 (f-Divergence)**.** Given a convex function $f : \mathbb{R} \to \mathbb{R}$ s.t. $f(1) = 0$, the f-Divergence is

$$D_f(p\|\pi) \equiv \mathbb{E}_p\left[f\left(\frac{p}{\pi}\right)\right] = \int p(x)\, f\left(\frac{p(x)}{\pi(x)}\right)\, dx \tag{4.13}$$

A common choice for $f := \log$; this choice is called the Kullback-Leibler (KL) Divergence

$$\mathsf{KL}(p\|\pi) := D_{\log}(p\|\pi) = \int p(x)\, \log\left(\frac{p(x)}{\pi(x)}\right)\, dx \tag{4.14}$$

It is preferred by computational staticians because it can be numerically estimated $\mathsf{KL}(p\|\pi) = \mathbb{E}_p[\log p] - \mathbb{E}_p[\log \pi]$

Property 2.
1. Not symmetric: $D_f(p\|\pi) \neq D_f(\pi\|p)$
2. Positive-definite: $D_f(p\|\pi) \geq 0$, and $D_f(p\|q) = 0 \iff p = q$

I am introducing this new family of divergences, instead of using $\mathsf{TV}(p,\pi) = \frac{1}{2}\int |p(x) - \pi(x)|\, dx$ because the TV is not differentiable.

Example. Let's compute the f-Divergence's time evolution under Langevin. Recall the law for Langevin for a target measure $\pi \propto e^{-U(x)}$ is

$$\partial_t p_t = -\nabla \cdot (\nabla U\, p_t) + \Delta p_t \tag{4.15}$$

$$
\begin{aligned}
\partial_t D_f(p_t \| \pi) &= \int \dot{p}_t(x)\, f\left(\frac{p_t(x)}{\pi(x)}\right) + \frac{p_t(x)\dot{p}_t(x)}{\pi(x)} f'\left(\frac{p_t(x)}{\pi(x)}\right) && (4.16)\\
&= \int \dot{p}_t(x) \left[f\left(r_t\right) + r_t f'\left(r_t\right)\right] \qquad\qquad r_t \equiv p_t/\pi && (4.17)\\
& && (4.18)
\end{aligned}
$$

If we specialize to the KL divergence, $f := \log$

Definition 8 (Log-Sobolev Inequality (LSI)). π satisfies a Log-Sobolev Inequality with constant $\lambda > 0$, if

$$
\forall p \in \mathcal{P}, \quad \mathsf{KL}(p\|\pi) \leq \frac{1}{2\lambda}\mathsf{FI}(p\|\pi) \tag{4.19}
$$

Fact (Bakery-Emery Criterion). Suppose that $\pi = e^{-U}$ with U β-strongly convex and λ-Lipschitz, that is $\lambda\mathbb{I} \preceq \nabla^2 f \preceq \beta\mathbb{I}$. Then π satisfies LSI with constant λ.

References

[Che26] Sinho Chewi. *Log Concave Sampling*. https://chewisinho.github.io/main.pdf.

[KE25] Volodymyr Kuleshov and Stefano Ermon. *Probabilistic Graphical Models Lecture Notes*. https://ermongroup.github.io/cs228-notes/

[Bru25] Joan Bruna. *Inference and Representation Lecture Notes*. https://joanbruna.notion.site/ir25

[Bru24] Joan Bruna. *Mathematics of Deep Learning Lecture Notes*. https://joanbruna.notion.site/Mathematics-of-DL-2c72610b1a9c401dbf624b3355fa0f2d

[Goo20] Jonathan Goodman. *Markov Chain Monte Carlo Lecture Notes*. https://math.nyu.edu/~goodman/teaching/MonteCarlo20/MonteCarloMethods.html

[Owe13] Art B. Owen. *Monte Carlo theory, methods, and examples*. https://artowen.su.domains/mc/

[LPW09] David A. Levin, Yuval Peres, and Elizabeth L. Wilmer. *Markov Chains and Mixing Times*. https://www.cs.cmu.edu/~15859n/RelatedWork/MarkovChains-MixingTimes.pdf